How the MLP Block Can Store a Fact

Contents

1. Overview . . . 2
2. Prerequisites . . . 3
3. The MLP block in one formula . . . 3
4. Rows ask questions; the activation makes an AND gate . . . 5
    4.1 Rows as questions . . . 5
    4.2 A toy fact: Michael Jordan plays basketball . . . 5
5. Columns write directions . . . 7
6. A real model: does GPT-2 small know sports? . . . 9
7. Where the sport comes from . . . 11
    7.1 Switching MLPs off . . . 11
    7.2 Who writes the answer at the last position? . . . 12
    7.3 The neurons . . . 14
8. What the toy leaves out . . . 15
9. Summary . . . 16
10. Sources . . . 16
11. Key terms . . . 17

Where this fits: Pipeline map step 8 (Model). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	**8 Model**	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Transformer encoder; Decoder-only GPT.

1. Overview

Key point: The **MLP block** (G-1245) of a transformer can be read as a long list of questions and answers: its first matrix asks, its second matrix writes the answer into the token's vector.

In more detail:

- The **MLP block** (multi-layer perceptron) is two dense layers, applied to one token's vector at a time.
- Each row of its first matrix asks a question about the token's vector (a **dot product** (G-634) plus a bias).
- The activation function (a function applied to each number; ReLU sets negatives to 0) keeps only clear "yes" answers, which makes a **neuron** (G-1317) behave like an AND gate.
- Each column of its second matrix is a direction that gets added to the vector when its neuron says yes.
- On GPT-2 small, which knows the sport of 18 of 19 athletes we tested, zeroing the early MLPs (the first blocks' feed-forward layers) at the athlete's name loses the right sport far more often than zeroing them anywhere else.

Give a language model the text "Michael Jordan plays the sport of" and it continues with "basketball". Somewhere in its weights the model holds that **fact** (G-745). Where, and in what form?

The feed-forward network (G-774), or MLP, was built from its shapes: two dense layers, applied to each token (a word or word piece) on its own. It holds about half of GPT-2 small's parameters (46 percent; GPT-2 small, matrix by matrix) and two thirds of GPT-3's (GPT-3), yet the shapes alone do not say what it does. This Note gives one reading of it, with a toy example and then a real model.

Figure 1 follows one toy neuron. Watch the number in the orange circle: it is 1 only when both names, Michael **and** Jordan, are in the input vector, and only then is the basketball direction added to the output.

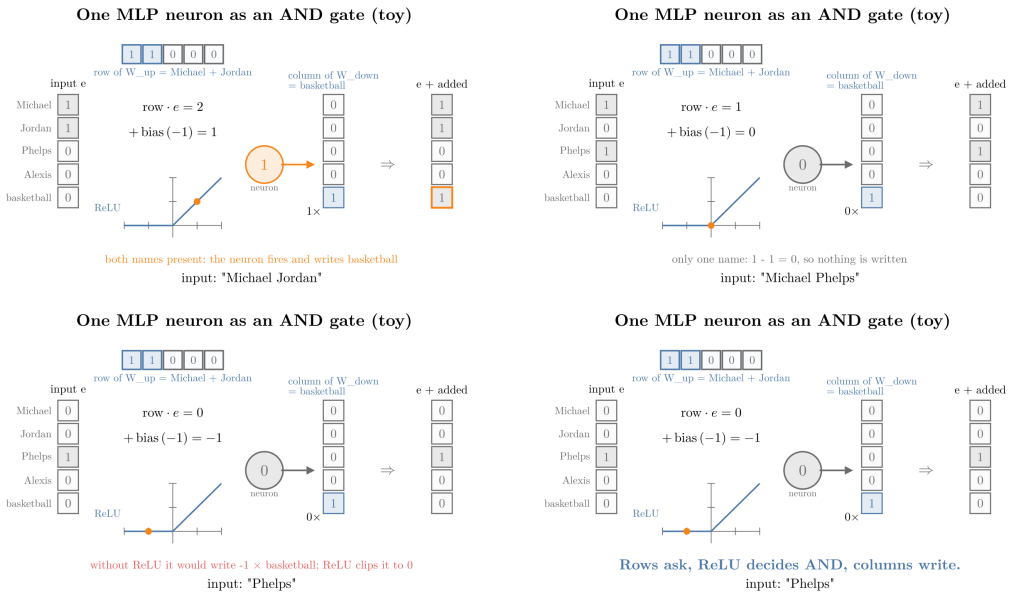


Figure 1: A toy MLP neuron. Input: a vector with 5 perpendicular directions. The neuron's row (blue) is Michael + Jordan, its bias −1, then ReLU. Its column (blue) is the basketball direction, added to the input when the neuron fires. Inputs in turn: Michael Jordan, Michael Phelps, Alexis Jordan, Phelps

The toy follows Sanderson (2024, Ch 7). The real-model part (sections 6 and 7) tests how much of the toy survives in GPT-2 small, and section 8 says what does not.

2. Prerequisites

- The feed-forward network: applied to each token separately, and its residual connection.
- Inside a GPT-2 block: GPT-2 small, the residual stream, **GELU** (G-834) and the parameter count.
- The unembedding: logits (the score for each token) as **dot products** (G-634) with token vectors; the logit lens.
- Arithmetic with directions and a dot product as a probe: directions in embedding space carry meaning; a dot product measures how much of a direction a vector has.
- Matrix multiplication as composition: a matrix times a vector as dot products with the rows, or as a sum of scaled columns.
- **ReLU** (G-1668): the activation that keeps positive **values** (G-2068) and sets negatives to 0.

3. The MLP block in one formula

> **Key point:** The **MLP block** (G-1245) takes one token's vector (a list of numbers), widens it, passes it through an activation function, narrows it back, and adds the result to the vector it started from.

We call the token's vector e, the widening matrix W_{up} with its bias b_{up}, and the narrowing matrix W_{down} with its bias b_{down} (G-42).

1. **In words:** the vector goes up to a wider space (4 times wider), through an activation function, back down to its own size, and is added to itself.
2. **Formula:**

$$u = W_{\text{up}}\, e + b_{\text{up}} \quad \text{(widen)}$$
$$a = f(u) \quad \text{(activation)}$$
$$d = W_{\text{down}}\, a + b_{\text{down}} \quad \text{(narrow)}$$
$$e \leftarrow e + d \quad \text{(add back)}$$

   All four lines in one, with the change d written out:

$$d = W_{\text{down}}\, f\Big(W_{\text{up}}\, e + b_{\text{up}}\Big) + b_{\text{down}}$$
$$e \leftarrow e + d$$

   where f is **ReLU** (G-1668) in the toy and GELU in GPT-2. (In GPT-2 a LayerNorm, which rescales the numbers, comes first, see prenorm; we leave it out here.)
3. **Example:** GPT-2 small: e has 768 numbers, W_{up} is $3{,}072 \times 768$, so the middle has 3,072 numbers (4×768), and W_{down} is $768 \times 3{,}072$.

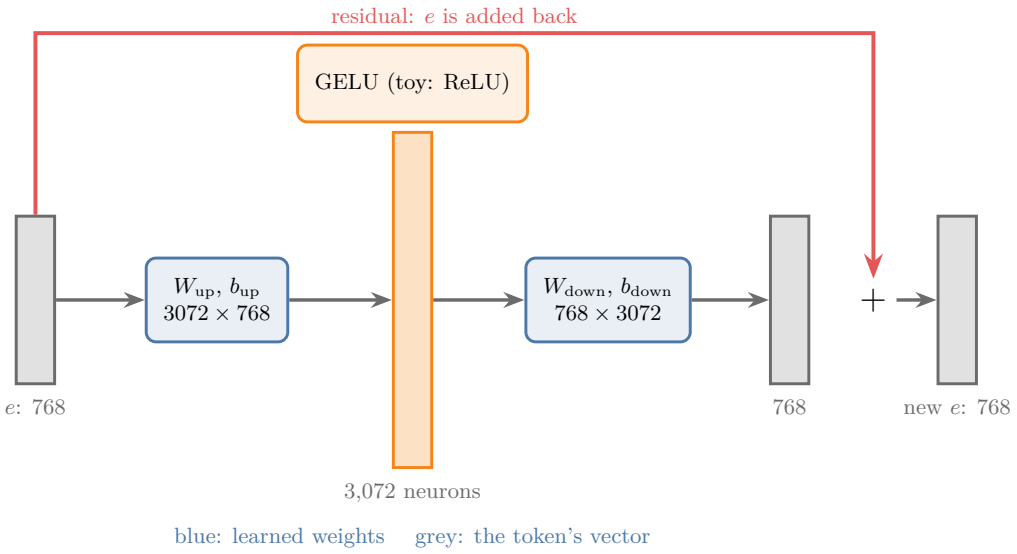


Figure 2: The MLP block of GPT-2 small with its shapes. The token's vector (grey, 768 numbers) goes up to 3,072 neurons through W_{up}, through the activation, back down to 768 numbers through W_{down}, and is added to the original vector (red)

Figure 2 shows the widths: the block widens each vector four times, then narrows it back so that it can be added to the stream.

The 3,072 middle **values** (G-2068) are the **neurons** (G-1317) of the block: "when you hear people refer to the neurons of a transformer, they're talking about these values" (Sanderson 2024, Ch 7). A neuron is **active** when its value is positive.

The block works on each token's vector alone, in parallel; tokens do not talk to each other here, only in attention (see the feed-forward network).

4. Rows ask questions; the activation makes an AND gate

> **Key point:** Each row of W_{up} is a direction, and its **dot product** (G-634) with e asks "how much of this direction does e have?". With the row Michael + Jordan and the bias -1, the answer is positive only when both names are present. **ReLU** (G-1668) clips every other answer to 0.

4.1 Rows as questions

> **Key point:** Multiplying by a matrix computes one dot product per row.

A matrix times a vector can be read in two ways (see computing a product column by column). The first: entry i of the result is the **dot product** (G-634) of row i with the vector. Here each row of W_{up} is a direction in the same 768-number space as e, so **neuron** (G-1317) i measures how well e points along row i. Sanderson (2024, Ch 7) calls each row a question asked about the vector.

4.2 A toy fact: Michael Jordan plays basketball

> **Key point:** With perpendicular directions M (first name Michael), J (last name Jordan) and B (basketball), a **neuron** (G-1317) with row $M + J$, bias -1 and column B adds basketball to "Michael Jordan" and to nothing else.

We follow Sanderson's toy (2024, Ch 7). Assume the vector space has perpendicular unit directions (arrows of length 1, at right angles to each other) for the ideas "first name Michael" (M), "last name Jordan" (J), "last name Phelps" (P), "first name Alexis" (A) and "basketball" (B). A vector "has" an idea when its **dot product** (G-634) with that direction is 1. The vector at the token " Jordan" in "Michael Jordan" then contains both names, $e = M + J$; attention has already brought "Michael" over from the previous token (Sanderson 2024, Ch 7).

1. **In words:** one neuron with row $M + J$ and bias -1, then **ReLU** (G-1668). Its column in W_{down} is B.
2. **Formula:**

$$n = \text{ReLU}\Big((M + J) \cdot e - 1\Big)$$

$$e \leftarrow e + n\,B$$

3. **Example:** for $e = M + J$, perpendicular unit vectors have dot product 0 and a unit vector with itself has 1:

$$(M + J) \cdot (M + J)$$
$$= M \cdot M + M \cdot J + J \cdot M + J \cdot J$$
$$= 1 + 0 + 0 + 1$$
$$= 2$$

Subtract the bias and apply ReLU:

$$2 - 1 = 1$$

$$n = \text{ReLU}(1) = 1$$

The output is $e + B = M + J + B$: it now also says "basketball".

The Notebook runs all the inputs of Figure 1:

Input e	Row $\cdot\, e$	+ bias -1	After ReLU	Basketball added
Michael Jordan, $M + J$	2	1	1	1
Michael Phelps, $M + P$	1	0	0	0
Alexis Jordan, $A + J$	1	0	0	0
Michael, M	1	0	0	0
Phelps, P	0	-1	0	0 (without ReLU: -1)

The bias sets the threshold: one name alone gives 0, so only both names together make the neuron positive:

$$1 - 1 = 0$$

The ReLU then clips every negative answer to 0. Without it, the input "Phelps" would write $-1 \times B$, an "anti-basketball" signal, into the vector. With it, the neuron outputs 1 for Michael **and** Jordan and 0 otherwise: it behaves like an **AND gate** (G-198) (Sanderson 2024, Ch 7).

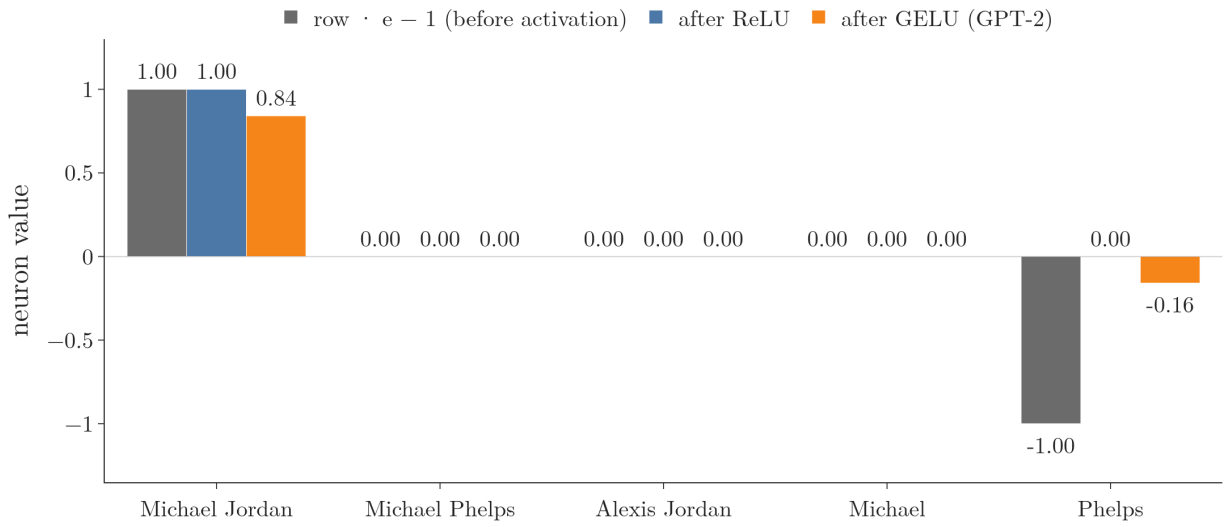


Figure 3: The toy neuron on the five inputs of the table: its value after the bias (grey), after ReLU (blue) and after GELU, GPT-2's activation (orange). Only "Michael Jordan" clears the threshold

Figure 3 puts the table's columns side by side. Grey shows the raw answer; ReLU removes the -1 of "Phelps"; GELU keeps a small negative **value** (G-2068) and lowers the "yes" to 0.84 (Extra below).

> **Extra:** GPT-2 uses GELU, $x\,\Phi(x)$ (Hendrycks and Gimpel 2016), instead of ReLU (**GELU** (G-834)). GELU has the same overall shape but is smooth: the toy neuron would give 0.84 for Michael Jordan instead of 1, 0 for one name, and -0.16 instead of 0 for "Phelps" (Notebook). The gate is softer but still separates the full name from the rest.

5. Columns write directions

> **Key point:** The output of W_{down} is a sum of its columns, each scaled by its **neuron** (G-1317)'s value. A column is a direction in the token's space: it is what that neuron adds to the stream when it fires.

The second reading of a matrix times a vector, also in the same section: the result is the sum of the columns, column i scaled by entry i of the vector. For W_{down} this reads

$$W_{\text{down}}\, n = n_1\, c_1 + n_2\, c_2$$

$$+ \cdots + n_{3072}\, c_{3072}$$

where c_i is column i, a 768-number direction. An inactive neuron ($n_i = 0$) adds nothing; an active one adds its column, scaled. In the toy, column 1 is B, so firing writes "basketball"

(Figure 1, right). A column can also carry several ideas at once, for example basketball plus other **facts** (G-745) about the same person (Sanderson 2024, Ch 7).

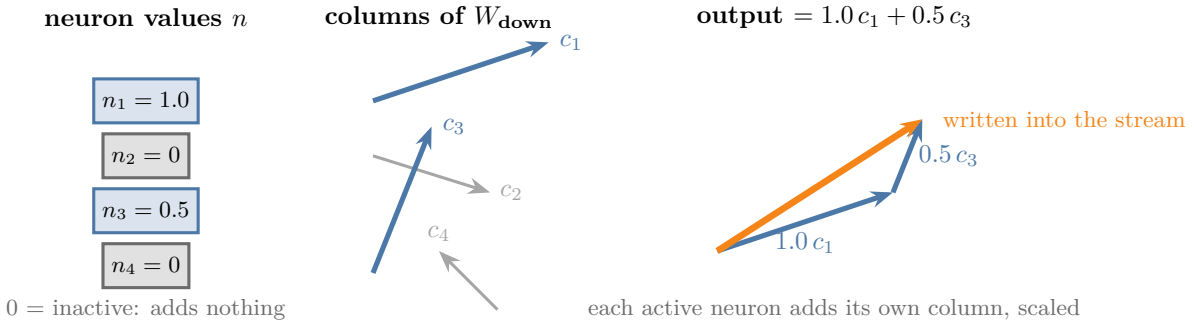


Figure 4: Columns write directions. Left: four neuron values, two active. Middle: the four columns of W_{down}, each a direction. Right: the output is $1.0\,c_1 + 0.5\,c_3$; the inactive neurons' columns add nothing

In Figure 4, follow the two blue arrows on the right: each active neuron adds its column, stretched by its value, and the orange sum is what the block writes into the stream.

Put together, the block is a store of question–answer pairs: 3,072 rows asking, 3,072 columns answering. Geva et al. (2021, §2, eq. 1) give the same reading and call the pairs **keys** (G-1011) and **values** (G-2068); in an MLP these are the **key and value** of G-1012. These keys and values are not the keys and values of attention; the names are reused. Geva et al. write the feed-forward layer, with x the token's vector, K the first matrix and V the second, as:

$$\text{FF}(x) = f(x \cdot K^T) \cdot V$$

- **Key:** a row of the first matrix. It "captures a particular pattern (or set of patterns) in the input sequence" (§2).
- **Value:** the matching row of the second matrix in their notation, our column. It "represents the distribution of tokens that follows said pattern" (§2): which tokens are likely to come next.

Geva et al. tested this on a 16-layer language model trained on WikiText-103 (a dataset of Wikipedia articles). They found keys whose top triggering text shared human-readable patterns: shallow ones (e.g. ending with "substitutes") in lower layers and more semantic ones in upper layers (Geva et al. 2021, §3, Table 1).

> **Extra:** The toy (Michael Jordan and basketball, the bias of -1), the AND-gate reading of **ReLU** (G-1668), the rows of W_{up} as questions and the columns of W_{down} as directions that get added follow Sanderson's *How might LLMs store facts* (3Blue1Brown, 2024, Ch 7). The Michael Phelps and Alexis Jordan inputs are his examples too. The animation's design and the numbers are our own.

6. A real model: does GPT-2 small know sports?

> **Key point:** Asked "⟨athlete⟩ plays the sport of", GPT-2 small ranks the right sport first for 18 of 19 athletes.

Before looking for a stored fact, we need a fact the model actually knows. Here a **fact** (G-745) is an athlete's sport, as the model's ranking of 10 sports after "⟨name⟩ plays the sport of". The Notebook asks GPT-2 small "⟨name⟩ plays the sport of" for 19 well-known athletes, the kind of fact studied by Nanda et al. (2023), and records the next-token probabilities:

Athlete	True sport	GPT-2's top token	Its probability
Michael Jordan	basketball	basketball	0.135
Kobe Bryant	basketball	basketball	0.617
Tiger Woods	golf	golf	0.615
Phil Mickelson	golf	football	0.071 (golf 0.049, second)
Wayne Gretzky	hockey	hockey	0.378
Roger Federer	tennis	tennis	0.540
Tom Brady	football	football	0.544
Michael Phelps	swimming	swimming	0.105
Sachin Tendulkar	cricket	cricket	0.415

(9 of the 19 shown; Figure 5 shows all.)

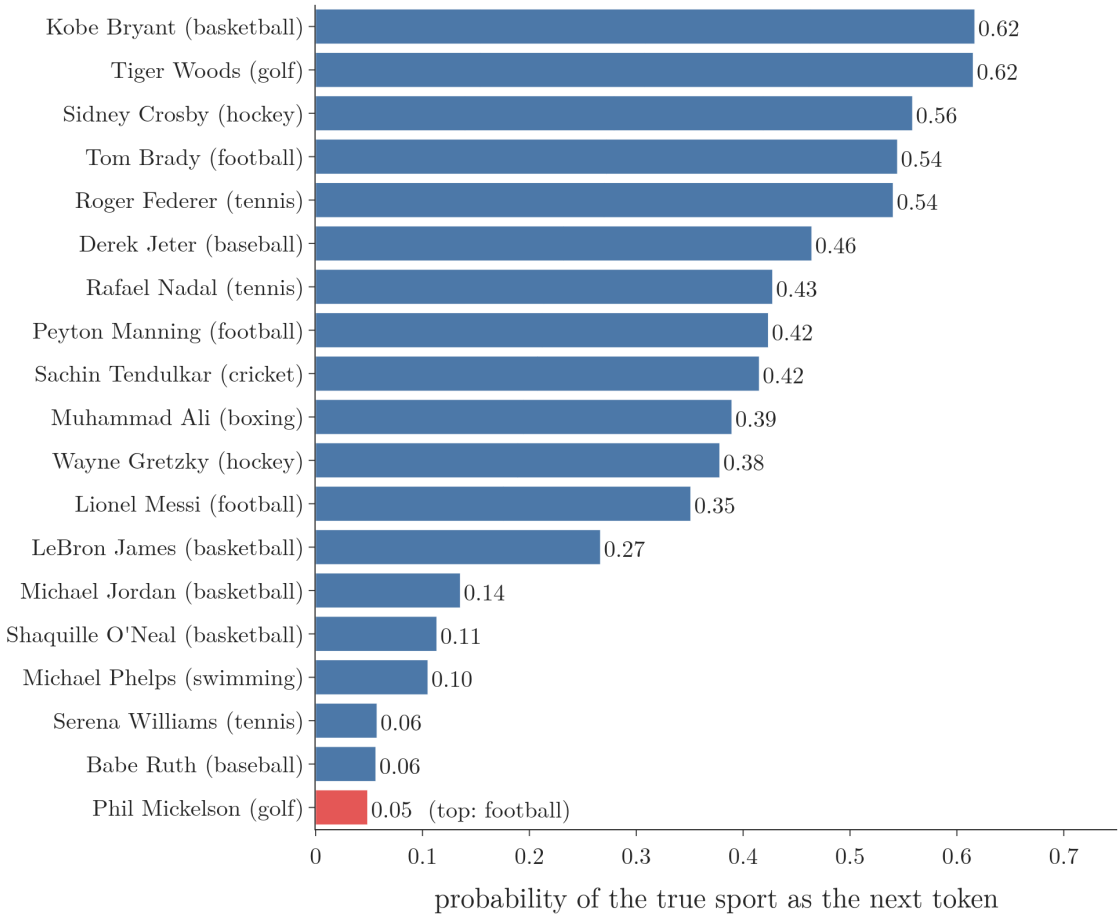


Figure 5: GPT-2 small's probability for each athlete's true sport as the next token after the prompt "[athlete] plays the sport of". Blue: the true sport is the top token; red: it is not

18 of 19 are right. Phil Mickelson is the one miss, so the experiments below use the other 18.

The probabilities can be low even when the answer is right: for Michael Jordan, " basketball" has 0.135, ahead of " football" at 0.086. Much of the probability goes to tokens that are not sports at all. The Notebook takes the 10 sports in the list (basketball, football, golf, hockey, tennis, baseball, soccer, cricket, boxing, swimming); for Michael Jordan, these 10 together get 0.415, so the rest goes elsewhere:

$$1 - 0.415 = 0.585$$

To measure only the fact, *which* sport, the Notebook asks whether the true sport is ranked first among the 10 sports, and computes its **share of the 10 sports**: its probability divided by the 10 sports' total. For Michael Jordan:

$$0.135/0.415 = 0.33$$

7. Where the sport comes from

> **Key point:** Zeroing the MLPs of the early blocks at the athlete's **name tokens** (G-1299) loses the right sport for many athletes; zeroing the same MLPs at the other tokens barely matters. The sport is then carried to the last position mainly by attention.

7.1 Switching MLPs off

> **Key point:** Zero MLP 1 at the name tokens and only 6 of 18 athletes keep the right sport first. Zero MLPs 2–6 there and 11 keep it; zero the same MLPs at the other tokens and 16 keep it.

To test what a group of MLPs does, we set their outputs to 0 at chosen positions and run the rest of the model unchanged (**zero-ablation** (G-2147)). The positions are either the **name tokens** (G-1299) (for example "Michael" and " Jordan") or the **other tokens** (" plays the sport of").

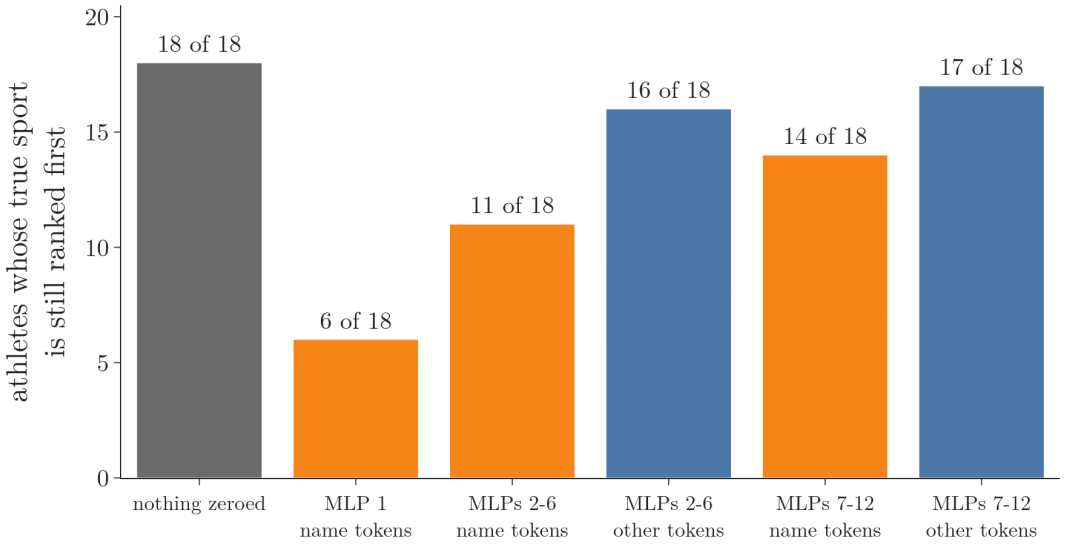


Figure 6: Athletes (of 18) whose true sport stays ranked first among 10 sports after zeroing a group of MLP outputs. Orange: zeroed at the name tokens. Blue: zeroed at the other tokens

MLPs zeroed	Where	True sport still first	Its share of the 10 sports	Its probability
none		18 of 18	0.62	0.36
block 1	name tokens	6 of 18	0.28	0.08
blocks 2–6	name tokens	11 of 18	0.44	0.13
blocks 2–6	other tokens	16 of 18	0.50	0.15

MLPs zeroed	Where	True sport still first	Its share of the 10 sports	Its probability
blocks 7–12	name tokens	14 of 18	0.49	0.25
blocks 7–12	other tokens	17 of 18	0.77	0.11

(Averages over the 18 athletes; Notebook.)

Three readings of Figure 6 and the table:

1. **The name tokens matter more than the rest.** For blocks 2–6, zeroing at the name tokens costs 7 athletes; zeroing at the other tokens costs 2. The same holds for blocks 7–12 (4 against 1).
2. **The early blocks matter more than the late ones** at the name tokens: block 1 alone costs 12 athletes, blocks 2–6 cost 7, blocks 7–12 cost 4.
3. **Late MLPs at the other tokens help with "a sport comes next", not with "which sport".** Zeroing them lowers the true sport's probability from 0.36 to 0.11, yet the right sport still ranks first for 17 of 18, and its share among the 10 sports even rises.

Nanda et al. (2023) found the same structure in a larger model, Pythia 2.8B (an open language model with 2.8 billion parameters), on athletes and their sports. Their circuit has three stages (Nanda et al. 2023, Post 1, summary of Post 2):

1. Attention layers "assemble the tokens of the athlete's name on the final name token": the last token of the name gathers all the name's tokens.
2. "MLPs 2 to 6 on the final name token map the concatenated tokens to a linear representation of the athlete's sport": these MLPs turn the gathered name into a direction in the vector that stands for the sport.
3. "A sparse set of mid to late attention heads extract the sport subspace and move it to the final token": a few attention heads in the middle and late blocks copy that sport direction to the last position. They describe the early MLPs as a "multi-token embedding": they turn a multi-token name into one vector from which attributes such as the sport can be read.

7.2 Who writes the answer at the last position?

> **Key point:** At the last token, the attention blocks together write most of the push towards basketball over football (1.22); the MLPs together add 0.23, with blocks 8 and 9 the largest.

The MLP works on each token alone, so whatever the MLPs wrote at the **name tokens** (G-1299) can reach the last position only through attention. We can measure what reached it. The final **residual stream** (G-1683; the token's vector that passes from block to block, each part adding its output) at the last token is a sum: the token-plus-position vector plus every attention and MLP output (see the residual stream grows). The measurement takes three steps (Notebook):

1. **The quantity.** The **logit difference** is the score of " basketball" minus the score of " football" at the last position: 0.454. A positive **value** (G-2068) means the model prefers basketball.
2. **Make it a sum.** The final LayerNorm divides the stream by its spread before the scores are computed. Freezing that divisor at its value in this run turns the step from stream to scores into plain multiply-and-add, so the score of a sum is the sum of the scores.
3. **Split it.** Each block's output then gets its own share of the logit difference, and the shares add up to 0.454 to three decimals.

For example, the two largest MLP shares (listed below) together push:

$$0.42 + 0.23 = 0.65$$

All shares together:

$$1.22 + 0.23 - 1.00 = 0.45$$

(attention, MLPs, and the final LayerNorm's shift).

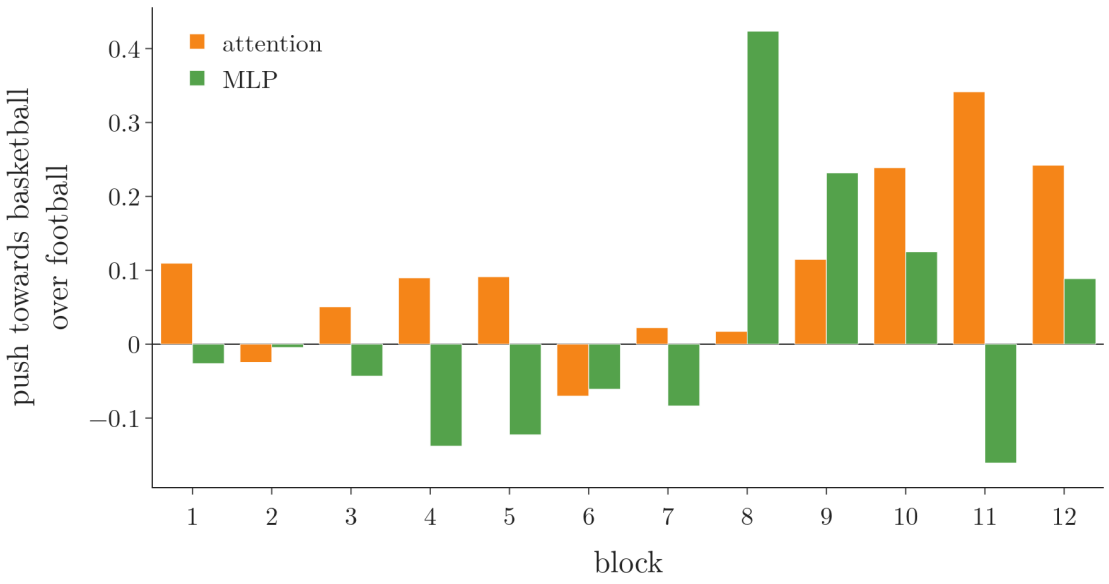


Figure 7: What each block's attention (orange) and MLP (green) added at the last position of "Michael Jordan plays the sport of", measured as a push towards " basketball" over " football"

- The 12 attention outputs together add +1.22; the 12 MLP outputs together +0.23; the final LayerNorm's shift adds −1.00.
- The largest single shares are MLP 8 (+0.42), attention 11 (+0.34), attention 12 (+0.24), attention 10 (+0.24) and MLP 9 (+0.23).

- Several MLPs push the other way: MLP 4 (−0.14), MLP 5 (−0.12), MLP 11 (−0.16).

This matches the third stage of Nanda et al. (2023): the sport is moved to the final token by attention. It also shows that the final answer is a sum of many small pushes in both directions, not one clean write.

7.3 The neurons

> **Key point:** Of the neurons in the three **MLP blocks** (G-1245) that push hardest (8, 9, 10), the neuron that pushes hardest writes a direction whose closest tokens are "championships, competitions, tennis, tournament": sport in general, not basketball.

Geva et al. (2021, §4) read a value vector by multiplying it with the output embedding and listing the tokens it scores highest. We do the same for the **neurons** (G-1317) that push hardest towards basketball over football at the last position (Notebook). "Its value" is the neuron's value (how strongly it fires); "Its push" is its share of the logit difference, the value times what its column adds to the basketball-minus-football score. A neuron with a large value can still push little:

Block	Neuron	Its value	Its push	Tokens its output direction scores highest
9	1704	1.59	0.165	championships, competitions, Championships, tennis, tournament, championship, tournaments, Olympic
10	992	0.95	0.069	calories, Exercise, Calories, workout, calorie, exercise, diet, diets
8	2957	2.34	0.050	poker, Dialogue, pond, rex, chess, vigil, rug, rel

Neuron 1704 of block 9 writes "competitive sport". Neuron 992 of block 10 writes "exercise". Neither writes "basketball". In block 8, the 10 strongest neurons give 71 percent of the block's push; in blocks 9 and 10 the 10 strongest give more than the whole block's push (201 and 230 percent), because other neurons push back. The answer is spread over many neurons.

8. What the toy leaves out

> **Key point:** The toy is a clean picture of what an MLP *can* do. Real models do not store a fact in one **neuron** (G-1317), and how they do store it is not fully known.

- **No single AND neuron.** Nanda et al. (2023) tested this directly: "We falsify the naive hypothesis that there's just a single step of detokenization, where the GELUs in specific neurons implement a Boolean AND on the raw tokens and directly output all known **facts** (G-745) about the athlete." Our neuron table points the same way: the strongest neurons write broad ideas, and many neurons share the work.
- **Not solved.** Sanderson (2024, Ch 7) introduces the DeepMind work with the caveat that "a full mechanistic understanding of how facts are stored remains unsolved". Nanda et al. (2023) "consider ourselves to have fallen short of the main goal of a mechanistic understanding of computation in superposition".
- **Directions need not be neurons.** The toy gave each idea its own perpendicular direction and each fact its own neuron. A real model has far more ideas than neurons or dimensions. Far more nearly perpendicular directions than dimensions fit into a space (see random directions are nearly perpendicular), and the **superposition hypothesis** (G-1916) is that models use this.

What does survive from the toy: the **MLP block** (G-1245) works on each token alone and adds its output to the stream; the early MLPs at the name tokens are where the sport becomes available (section 7.1); and the reading of rows as pattern detectors and outputs as written directions is the one Geva et al. (2021) support with evidence.

9. Summary

Part	Toy reading	What GPT-2 small shows
Row of W_{up} + bias	a question: "Michael **and** Jordan?"	3,072 rows per block
Activation	ReLU makes an AND gate	GELU, a softer gate
Column of W_{down}	the direction written: basketball	strongest neurons write "competitive sport", "exercise"
Where the fact appears	in the MLP at the name token	zeroing MLPs 1–6 at the name tokens loses the sport far more often than elsewhere
How it reaches the answer	(not in the toy)	mostly attention at the last token

- The **MLP block** (G-1245) widens each token's vector, applies the activation, narrows it back and adds the result to the vector (formula in section 3), for each token on its own, so tokens do not exchange information here, only in attention.
- Rows ask questions by dot products; the bias sets a threshold; the activation keeps only clear "yes" answers, so one **neuron** (G-1317) can fire for Michael **and** Jordan and stay at 0 for either name alone.
- Columns are directions added to the vector, scaled by their neuron, so an inactive neuron writes nothing and an active one writes its idea into the stream.
- GPT-2 small knows 18 of 19 athletes' sports; the early MLPs at the **name tokens** (G-1299) matter most for which sport (zeroing MLP 1 there leaves only 6 of 18 right); attention carries the answer to the last token, because the MLP works on each token alone.
- Real **facts** (G-745) are spread over many neurons, and the full mechanism is still unknown, so the toy AND neuron is a picture of what an MLP can do, not of how GPT-2 stores a fact.
- So the answer to "where is the fact?" is: largely in the early MLPs at the name tokens, written as directions, and spread over many neurons rather than one.

10. Sources

Built from

- Sanderson, G. (3Blue1Brown), "How might LLMs store facts | Deep Learning Chapter 7", 2024, 3blue1brown.com/lessons/mlp, https://www.youtube.com/watch?v=9-Jl0dxWQs8
- Geva, M., Schuster, R., Berant, J. and Levy, O. (2021). Transformer Feed-Forward Layers Are Key-Value Memories. *EMNLP 2021.* arXiv:2012.14913 (v2). Abstract; §2 (eq. 1, $\text{FF}(x) = f(x \cdot K^T) \cdot V$; keys capture patterns, values represent the distribution of following tokens); §3 and Table 1 (human-readable key patterns; the 16-layer model of

Baevski and Auli trained on WikiText-103); §4 (value vectors read through the output embedding, $p = \text{softmax}(v \cdot E)$).

Other references

- Nanda, N., Rajamanoharan, S., Kramár, J. and Shah, R. (2023). Fact Finding: Attempting to Reverse-Engineer Factual Recall on the Neuron Level (Post 1). AI Alignment Forum, 23 December 2023, alignmentforum.org/posts/iGuwZTHWb6DFY3sKB. Executive summary (Pythia 2.8B, athletes and 3 sports; the three-stage circuit; "multi-token embedding"; the falsified single-neuron AND hypothesis; "fallen short of the main goal").
- Hendrycks, D. and Gimpel, K. (2016). Gaussian Error Linear Units (GELUs). arXiv:1606.08415. §2 (GELU).
- Weights and tokenizer: `openai-community/gpt2`, huggingface.co.

11. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
MLP block (G-1245)	The feed-forward part of a transformer block: it widens each token's vector with a matrix (up-projection), applies an activation, shrinks it back (down-projection) and adds the result to the token's vector.
Fact (here) (G-745)	An athlete's sport, as the model's ranking of 10 sports after "⟨name⟩ plays the sport of".
W_{up}, W_{down} (G-42)	The two learned matrices of the MLP block: W_{up} first widens each vector, W_{down} then narrows it back. In GPT-2 they are named `c_fc` and `c_proj`.
AND gate (G-198)	A rule that outputs "yes" only when all its inputs are "yes".
Key, value (in an MLP) (G-1012)	Geva et al.'s names for a row of the first matrix (a pattern detector) and the matching direction of the second matrix (what gets written).
Name tokens (G-1299)	The tokens that spell the athlete's name in a prompt such as "Michael Jordan plays the sport of"; zeroing the MLP outputs at these positions tests whether the MLPs there recall the sport.

Term	Meaning
Zero-ablation (G-2147)	Setting a part's output to 0 and running the rest of the model unchanged, to see what that part does.
Dot product (G-634)	Multiplying two vectors entry by entry and adding the products, $u^{\mathsf{T}}x$, which gives one number. It is large when the vectors point the same way, so it measures similarity and gives projections.
Neuron (in a network) (G-1317)	One unit of a neural network (in an ANN, a perceptron): it takes inputs from the layer before, combines them with its learned weights and passes its output on.
Feed-forward network (FFN) (G-774)	A small two-layer network inside each transformer block that works on each word's vector on its own: dense layers $512 \to 2048$ with ReLU $\to 512$. Attention mixes the words; this network transforms each word and adds the block's only per-word non-linearity.
GELU (G-834)	A smooth activation function that multiplies each input x by $\Phi(x)$, the chance that a standard normal value is below x: $x\,\Phi(x)$ (Gaussian Error Linear Unit). GPT's MLPs use it.
ReLU (G-1668)	Rectified linear unit, the activation $\max(0, z)$: it passes a positive input unchanged and turns a negative one into 0; the usual default activation for hidden layers.
Value (G-2068)	In self-attention, the vector each word contributes to the weighted sum that becomes a word's new vector; it comes from the word's embedding through a learned value matrix.
Key (G-1011)	In attention, the vector each word offers to be compared with a query; the dot product of the query with a key scores how much attention that word gets.
Residual stream (G-1683)	The token vector that passes from block to block, to which every sub-layer adds its output.
Superposition hypothesis (G-1916)	The proposal that real neural networks store features in superposition (Elhage et al. 2022).